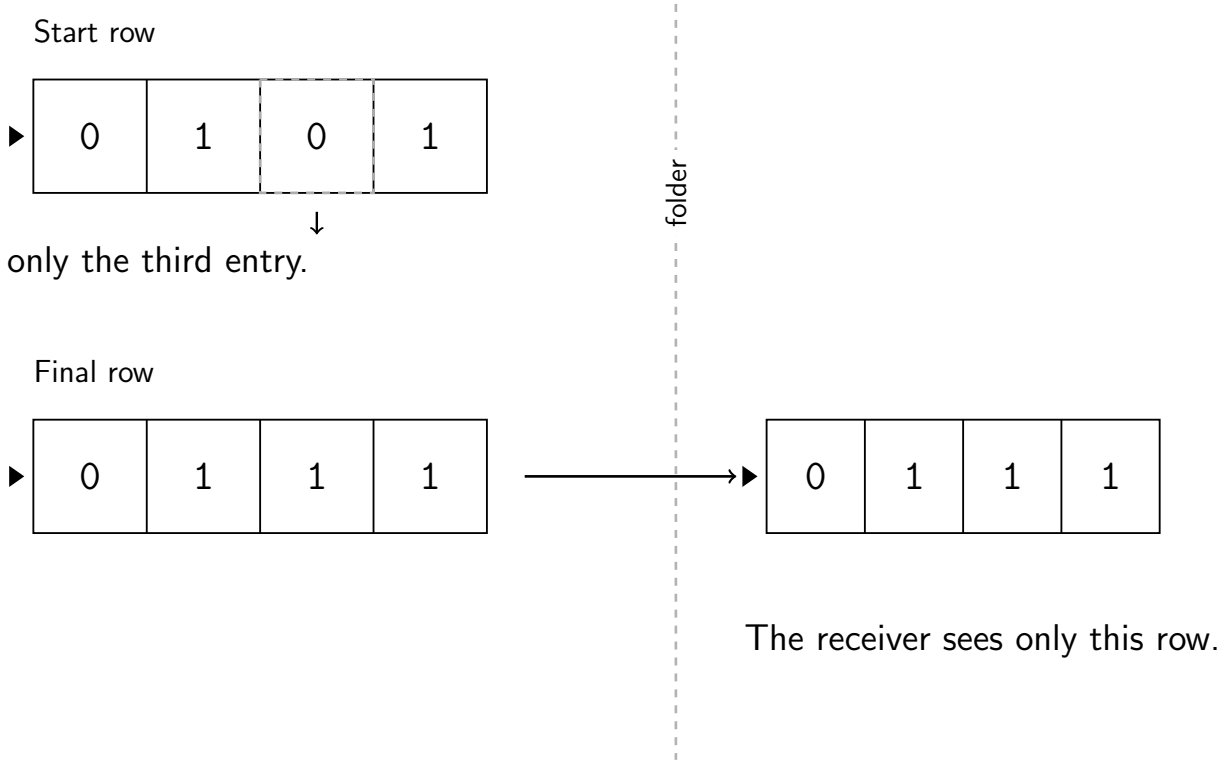
Use 0 for an empty circle and 1 for a filled circle. A key gives a different row to each picture, all the same length. The sender hides the picture and its row. A changer changes zero or one entry, without saying where or whether a change occurred. The receiver gets only the final row and knows the key. Keep the arrow at the left.

Example: a practice row

Sender and changer

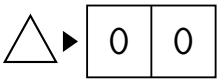
Receiver

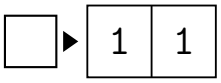


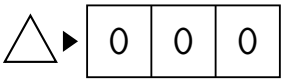
In the game, the start row and the changed position stay hidden behind the folder.

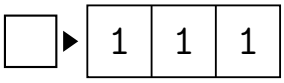
The changer may also leave the row as it is.

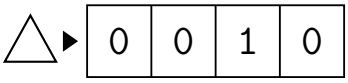
Problem 1: For each key, find every row the receiver might see. Mark any row that could mean either picture. Which keys always let the receiver recover the picture?

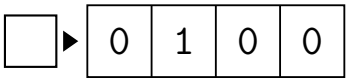
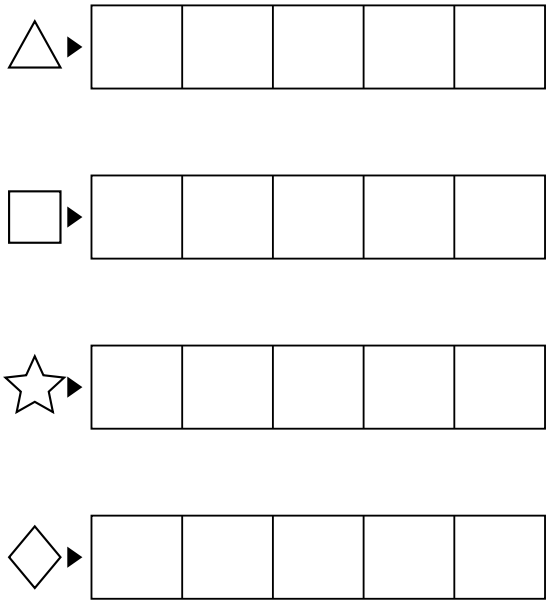












Problem 2: Make a key for all four pictures with five entries in each row. The receiver must always recover the picture after zero or one change. Explain why your key works.